# Air Pressure Dependence of the Speed of Light in Air Measured with the Michelson Interferometer

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We measured how the speed of light in air depends on the surronding air pressure using a Michelson interferometer with a vacuum cell in one arm. While the cell was pumped down from atmospheric pressure to 36.5 kPa, we counted the fringes passing a fixed reference point on the screen and recorded the pressure at each fringe passing through the point. Three sweeps of 15 points each gave consistent slopes, with a combined value of $-0.2148(13)\,\mathrm{kPa}^{-1}$. Using Eq. (3) [6, Eqs. (3)], this gives a pressure coefficient $\beta = 2.09(3) \times 10^{-6}\,\mathrm{kPa}^{-1}$ and a speed of light in air of $2.997\,28(1) \times 10^{8}$ m/s at 103.2 kPa, which is 64.7(11) km/s below the vacuum value. The fringe count is linear in pressure within the counting uncertainty, although the residuals show a slight curvature. Notably we used the measured outer length for the vaccum cell length, while the actual air path is likely shorter, which results in a bias in the result.

## INTRODUCTION

In 1849, Fizeau made the first terrestrial measurement of the speed of light, sending light through several kilometres of air [1]. Light in air is slightly slower than in vacuum, but the difference is only about a few parts in $10^4$, so measuring it requires a very sensitive method. The Michelson interferometer is such a method, which turns a tiny change in light-travel time into a visible motion of interference fringes. Michelson and Morley used it in 1887 in their search for the ether, and their null result made interferometry a standard tool for precision optical measurements [2].

The speed of light in air also matters in practice. Satellite navigation, ranging and precise timing all rely on signals that travel through the atmosphere. The neutral atmosphere delays these signals, and this delay depends strongly and predictably on pressure, so it must be modelled to obtain accurate positions [3].

In this experiment we use a tabletop Michelson interferometer equipment with an vaccum cell in one arm and count the fringes that pass a fixed point as the cell pressure is lowered from 103.2 to 36.5 kPa. We test whether the fringe count depends linearly on pressure, determine the pressure coefficient $\beta$ from three independent sweeps, and infer $c_{\mathrm{air}}(P)$ over the measured range.

## METHODS

Measurements were made with a PASCO OS-9255A Michelson interferometer [4] illuminated by a He–Ne laser ($\lambda_0 = 632.8$ nm). A vacuum cell was placed in the arm of the movable mirror $M_2$. Before the experiment we adjust the equipments, first we removed the diverging lens and overlapped the two returning beam spots on the viewing screen; the lens was then inserted, and then by turning the $M_2$ micrometer we confirmed fringe moved correctly. The cell faces were oriented perpendicular to the beam. While The fringes were in an oval shape rather than perfectly circular, probably due to tiny misalignment of the equipments. The outer length of the cell was measured as $L = 3.25(5)$ cm; because the tube is has a completely black shell, the positions of the windows could not be determined precisely.

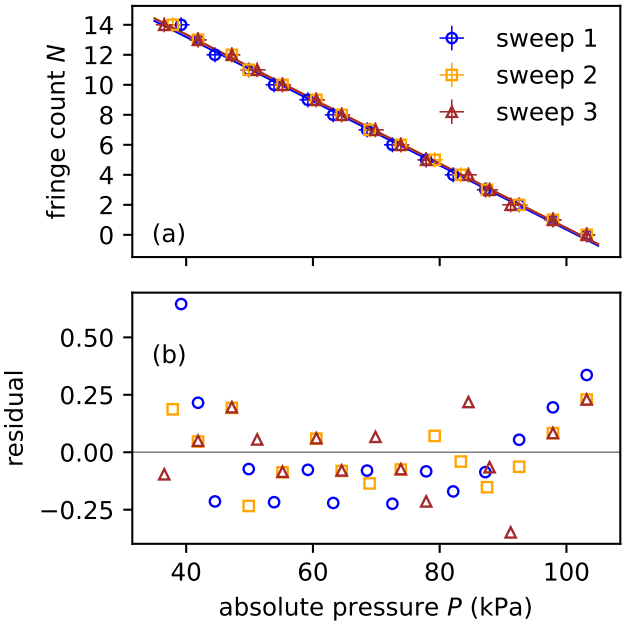



FIG. 1. (a) Cumulative fringe count $N$ versus absolute pressure $P$ for three independent sweeps (circles, squares, triangles), recorded while pumping the cell down from atmospheric pressure. Lines are least-squares fits $N = mP + b$ to each sweep, with slopes $m = -0.214(3)\,\mathrm{kPa}^{-1}$, $-0.2150(18)\,\mathrm{kPa}^{-1}$ and $-0.215(2)\,\mathrm{kPa}^{-1}$; their weighted mean is $\bar{m} = -0.2148(13)\,\mathrm{kPa}^{-1}$. Error bars show the estimated counting uncertainty (0.5 fringe) and gauge-reading uncertainty (1.33 kPa). (b) Residuals $N - (mP + b)$ for each sweep. All three sweeps show a slight curvature, positive at both ends, with most residuals within $\pm 0.25$ fringe.

A reference mark was placed on the screen and the cumulative fringe count was set to $N = 0$ at atmo-

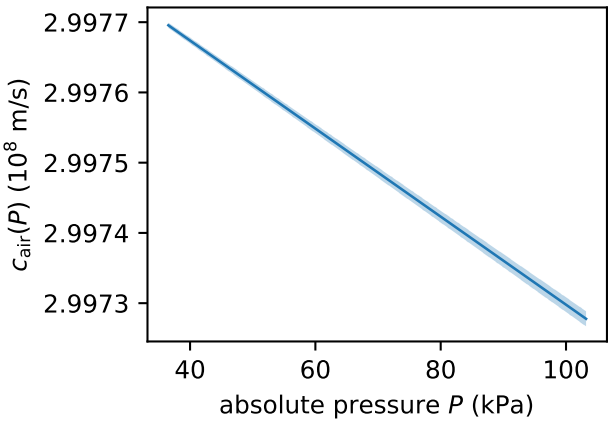



FIG. 2. Speed of light in air $c_{\mathrm{air}}(P)$ over the measured pressure range, calculated from Eq. (3) using the combined slope $\bar{m}$ and the measured cell length $L = 3.25\,\mathrm{cm}$. The shaded band shows the one-standard-deviation uncertainty from $\bar{m}$ and $L$. At atmospheric pressure, $c_{\mathrm{air}} = 2.997\,28(1) \times 10^8\,\mathrm{m/s}$, which is $64.7(11)\,\mathrm{km/s}$ below the vacuum value. Because $L$ is the outer length of the cell, $c_{\mathrm{air}}(P)$ is likely overestimated(see Methods).

spheric pressure. While the cell was pumped down, the gauge reading $P_g$ (in cmHg, relative to atmosphere) was recorded each time a bright fringe crossed the mark, giving 15 points per sweep down to $P_g = 48$–$50\,\mathrm{cmHg}$, which close to the gauge limit. One trial sweep(not used in the analysis) and three sweeps were taken, all while pumping down. Gauge readings were converted to absolute pressure with $P = P_{\mathrm{atm}} - P_g$, using $P_{\mathrm{atm}} = 103.20(5)\,\mathrm{kPa}$ for the day of the measurement, giving a range of 36.5–103.2 kPa.

Each sweep was individually fitted with a straight line $N = mP + b$ by least squares method, with the slope uncertainty estimated from the scatter of the residuals [5, Eqs. (5.1)–(5.6)]. The three slopes are consistent within 0.3 of their combined uncertainty and were combined by a weighted mean [5, Sec. 4.5], giving $\bar{m} = -0.2148(13)\,\mathrm{kPa}^{-1}$. Because $N$ increases as the pressure decreases, the sign of $\bar{m}$ was reversed to match the convention of Eq. (1) before applying Eq. (3) [6, Eqs. (1)(3)]. The analysis was performed in Python with NumPy and Matplotlib [7].

The estimated reading uncertainties were $\sigma_N = 0.5$ fringe and $\sigma_{P_g} = 1\,\mathrm{cmHg}$; the scatter of the residuals, 0.14–0.26 fringe, is smaller than estimated. The residuals show a slight curvature common to all three sweeps (positive at both ends of the pressure range and negative in the middle). They do not exceed about 0.6 fringe and most lie within $\pm 0.25$ fringe, below the counting uncertainty, so the linear model is adequate at this level of precision; the curvature may indicate a systematic effect, such as pressure changes during gauge readings. The reverse way of pumped down the cell was not tested. Among the statistical uncertainties, that of $L$ dominates over that of the slope.The PASCO manual gives the actual length of the vacuum cell as 2.54 cm [4], so $L$ is in fact the largest source of error in this experiment.

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